

YUZHUAO JIA

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EDUCATION

Nanfang College of Sun Yat-sen University

B.Sc. Computer Science and Technology. GPA: 3.2/4.0

Thesis: Job Position Data Analysis Based on Web Crawling Techniques

Guangzhou, China

Sep 2017 – Sep 2021

University of Sydney

M.Sc. Computer Science. GPA: Credit

Thesis: Teaching DCD children how to move, using Augmented Reality(AR)

Sydney, Australia

Feb 2023 – Feb 2025

RESEARCH INTERESTS

My research interests lie at the intersection of Agentic AI, Large Language Models, and Human-Computer Interaction, with a particular focus on adaptive interfaces, embodied cognition, and AI-driven learning systems. My previous work spans designing adaptive reading systems with real-time gaze and facial sensing for middle-aged users, developing AR-based therapeutic tools for children with Developmental Coordination Disorder (DCD), and building voice-based interactive learning systems for early childhood math education. I also have experience in fine-tuning LLMs for domain-specific applications, as well as medical imaging narrative generation and remote sensing segmentation. Currently, my work focuses on embodied cognition and AI, specifically investigating whether LLMs can develop human-like bodily representations when interacting with virtual environments.

PUBLICATIONS

Shu, K. *, **Jia, Y. ***, Zhang, Z., & Gao, J. (*Equal contribution). (2024). *FODAP Graph for Enhanced Medical Imaging Narrative Generation*. Accepted at International Conference on Bioinformatics and Biomedicine (BIBM). DOI: 10.1109/BIBM62325.2024.10822532

Yu, K. *, **Jia, Y. ***, Fan, H., Zheng, C., & Peng, Z. (*Equal contribution). (2026). *Balancing Automation and Agency: Designing Adaptive Reading Interfaces for Middle-Aged Office Workers*. Accepted at ACM CHI Conference on Human Factors in Computing Systems (CHI 26), Poster.

Mao, Y. *, Li, Y. *, **Jia, Y.**, Li, F., Yin, Z., & Zheng, S. (*Equal contribution). (2026). *LUMOS: Designing an Interactive Companion Robot to Enhance Critical Reading and Thinking in Academic Contexts*. Accepted at the 2026 35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).

Jia, Y. (2024). *Pedestrian Behavior Detection and Traffic Violation Recognition Based on YOLOv5*. Accepted at the 2024 4th International Conference on Image Processing and Intelligent Control. DOI: 10.1117/12.3038591

Jia, Y. *BiFocalNet: Dual-Branch Architecture for Enhanced Remote Sensing Segmentation*. Under revision at IEEE Transactions on Geoscience and Remote Sensing.

Anonymous Author(s) (as the first author). (2026). *Human and LLMs' bodily representations*. Preparing for Nature Machine Intelligence.

RESEARCH EXPERIENCE

Tsinghua University

Research Assistant

Beijing, China

Human and LLMs' Bodily Representations Research

Sep 2025 – Present

- Investigating whether pretrained Large Language Models can develop human-like bodily representations when interacting with virtual environments, addressing a critical gap in embodied cognition research across the spectrum from conceptualization to sensorimotor experience.
- Participated in designing experimental paradigms for comparing human and LLM cognitive representations on embodied tasks, conducting literature review across top-tier journals (Nature Machine Intelligence, Nature Human Behavior, PNAS, Cognition) to inform task selection.

- Developed computational analysis framework including model selection and deployment of open-source vision-language models, layer-wise embedding extraction and analysis, dimension reduction via Principal Component Analysis (PCA), t-SNE clustering for representation visualization, and Representational Similarity Analysis (RSA) to quantify representation overlaps and differences between humans and LLMs.
- Contributing to manuscript preparation for submission to top-tier cognitive science journals, with responsibilities in experimental design, computational modeling, data analysis pipeline development, and writing Methods and Results sections.

Adaptive Reading System

Jun 2025 – Sep 2025

- Architected and implemented a browser-based adaptive reading system as a lightweight Tampermonkey userscript. The core sensing module leverages MediaPipe Face Mesh for real-time facial landmark detection, enabling precise viewing distance calculation to dynamically adapt font size and color contrast.
- Executed a comprehensive user study with 24 middle-aged adults, evaluating the system's impact on reading speed, comprehension accuracy, and comfort across four distinct reading conditions.
- Performed qualitative data analysis on semi-structured interviews using thematic analysis, uncovering key usability insights, such as the trade-off between the cognitive load of adaptations and performance gains.
- Co-authored the research paper, contributing to the literature review (Related Work), system design details (System Design), and interpretation of findings (Discussion).

Embodied Intelligence Research Group

Apr 2025 – Aug 2025

- Engineered and constructed a 6-DOF force-controlled robotic arm utilizing 3D-printed components and DAMIAO motors, currently integrating the system with MoveIt 2 for advanced motion planning and control.
- Extended an open-source framework to develop an enhanced web-based agentic system with speech-to-text (STT) capabilities, real-time video streaming functionality, and Model Context Protocol (MCP) client integration, enabling seamless compatibility with all MCP-compliant tools and services.

PI Lab – MathAdventure

Jan 2025 – Mar 2025

- Contributed to MathAdventure, a novel screen-free, voice-based interactive learning system grounded in Contextual Learning and Zone of Proximal Development (ZPD) principles, designed to enhance mathematical understanding of the decimal system in children ages 4-8 through real-world object symbolization.
- Built the core computer vision module for real-time object recognition (e.g., cups, spoons), enabling contextualized mathematical problem generation based on children's immediate physical surroundings—a key aspect of the project's pedagogical design.
- Orchestrated comprehensive user studies including Wizard of Oz experiments and iterative prototype testing with 18 child-parent pairs, performing structured interviews and ensuring methodologically sound experimental procedures to evaluate engagement, learning effectiveness, and ZPD-based scaffolding strategies.
- Authored significant sections of the manuscript "MathAdventure: Contextual and Zone of Proximal Development-driven Math Learning through Symbolizing Real-World Objects with the Decimal System", including the Related Work and key portions of System Design and Development.

University of Sydney

Sydney, Australia

BiFocalNet: Dual-Branch Architecture for Enhanced Remote Sensing Segmentation

Jun 2024 – Dec 2024

- Proposed BiFocalNet for remote sensing segmentation, achieving SOTA on GID with +3.06 mIoU over prior work.
- Designed a dual-branch encoder (EfficientNetV2 + PVT) to capture local details and global context, plus Cross-Fusion and SuperASPP for feature fusion and multi-scale modeling.
- Ran extensive experiments/ablations; improved over ParaTransCNN by +3.28 (Forest) and +3.51 (Built-up) mIoU.

FODAP Graph for Enhanced Medical Imaging Narrative Generation

Mar 2023 – May 2024

- Built a MedSAM-based ViT visual encoder for 512×512 medical images to generate high-quality representations.
- Simplified the encoder by removing the MLP neck and using patch embeddings directly; reduced features from 1024×768 to 256×768 to unify outputs and improve generalization.
- Benchmarked against BioMedCLIP-pretrained ViT-B/16@224 and ImageNet21k-pretrained CvT@384, showing MedSAM's best performance.

PROJECTS

Fine-tuning LLMs for Struggling Student Simulation | [GitHub](#)

Dec 2025 – Jan 2026

- Designed a two-stage LLM augmentation pipeline using MisstepMath and Gemini-3-Pro API, generating 1,300+ student misconception samples with structured error type and metacognitive annotations.
- Applied LoRA fine-tuning on Qwen3-8B to simulate authentic struggling learners, training only 0.059% of parameters while achieving 100% metacognitive deficit behaviors and 37.5% realistic error production.
- Validated instruction-tuning approach for educational AI, enabling scalable development of intelligent tutoring systems and teacher training programs.

Interview Assistant | [GitHub](#)

Sep 2024 – Nov 2024

- Developed a cross-platform (Windows/macOS) desktop application using Electron to provide real-time, AI-driven response suggestions during online interviews.
- Architected a real-time audio processing pipeline integrating Deepgram for low-latency speech-to-text and OpenAI's GPT for generating context-aware suggestions, achieving an average response time of under 2 seconds.
- Implemented a key personalization feature allowing users to upload their own documents (resume, project details), enabling the GPT model to generate highly tailored answers reflecting the user's unique experience.
- Ensured user privacy by processing all personal data locally and managing API interactions securely.

Teaching DCD Children How to Move Using Augmented Reality (AR)

Aug 2024 – Dec 2024

- Developed a therapeutic web-based AR application using WebXR to guide children with Developmental Coordination Disorder (DCD) through motor skill exercises.
- Engineered a real-time motion analysis and scoring system by integrating the MediaPipe pose estimation model; the system precisely tracks and evaluates key indicators like arm height and trunk stability to provide immediate, actionable feedback.
- Designed and implemented a child-centric UI/UX in Figma, featuring a gamified, positive reinforcement scoring system that was shown in user testing to significantly boost engagement and confidence among children.
- Built interactive 3D training modules, allowing users to practice complex movements (e.g., balancing, throwing) in a safe, guided AR environment with voice and text prompts.

AWARDS & ACTIVITIES

Excellence Award: National University Student Knowledge Competition. (2018)

Academic Excellence Scholarship: Awarded for ranking in the top 5% of the academic cohort. (2020)

Director, Secretariat Department, Student Union: Directed resource & budget management and orchestrated key campus-wide events like the Student Congress. (2018 – 2019)

Director, Planning Department of Japanese Language Association: Led a team to plan and execute cultural events to boost member engagement. (2019 – 2020)

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, LaTeX

Machine Learning / CV: PyTorch, MediaPipe, ROS 2

Web & Application Dev: React, Next.js, Electron, WebXR, Tailwind CSS, Node.js

Tools & Platforms: Git, Unity, Arduino, SQL Databases, Figma

Research & Design: UI/UX Prototyping, Qualitative & Quantitative Analysis